ACL

1 ACL Model

ACL is the shared organizational and domain-state language used by iStar and BPMN. An ACL model is

$$\mathcal{A} = \langle N, \mathit{Class}, \mathit{Entity}, \mathit{Role}, \mathit{OrgCtx}, \mathit{Att}, \mathit{Assoc}, \mathit{associates}, \mathit{rolenames}, \mathit{multiplicities}, \prec, \mathit{Inv} \rangle.$$

The class categories are finite and pairwise disjoint:

$$\mathit{Class} = \mathit{Entity} \uplus \mathit{Role} \uplus \mathit{OrgCtx} \subseteq N.$$

Entity represents domain concepts, *Role* organizational concepts, and *OrgCtx* the organizational context in which the shared state is interpreted.

2 Attributes and Associations

For every $c \in \mathit{Class}$,

$$\mathit{Att}_c \subseteq \{a : t_c \to t \mid a \in N,\ t \in T\}.$$

Associations are finite names $\mathit{Assoc} \subseteq N$ with

$$\mathit{associates} : \mathit{Assoc} \to \mathit{Class}^+, \qquad \mathit{rolenames} : \mathit{Assoc} \to N^+,$$

and, whenever $\mathit{associates}(as) = \langle c_1, \dots, c_n \rangle$,

$$|\mathit{rolenames}(as)| = n \geq 2.$$

Multiplicities are

$$\mathit{multiplicities} : \mathit{Assoc} \to \big(\mathcal{P}(\mathbb{N}_0) \setminus \{\varnothing\}\big)^+,$$

with one multiplicity per association end.

3 Generalization

Generalization is type preserving:

$$\prec \subseteq (\mathit{Entity} \times \mathit{Entity}) \cup (\mathit{Role} \times \mathit{Role}) \cup (\mathit{OrgCtx} \times \mathit{OrgCtx}).$$

Let $\preceq$ be its reflexive-transitive closure. For Entity and OrgCtx, specialization uses domain inclusion:

$$c \prec c' \wedge c, c' \in \mathit{Entity} \cup \mathit{OrgCtx} \Longrightarrow I_{\mathit{Class}}(c) \subseteq I_{\mathit{Class}}(c').$$

For Role, role instances remain distinct; a specialization is represented by an explicit play/generalization relation in a state rather than by object identity.

4 Potential Object and Link Domains

For each class c, let $oid(c)$ be an infinite set of potential identifiers. Define

$$I_{Class}(c) = \begin{cases} \bigcup_{c' \preceq c} oid(c'), & c \in Entity \cup OrgCtx, \\ oid(c), & c \in Role. \end{cases}$$

If $associates(as) = \langle c_1, \ldots, c_n \rangle$, then

$$I_{Assoc}(as) = I_{Class}(c_1) \times \cdots \times I_{Class}(c_n).$$

5 ACL State

An ACL state is

$$\sigma = \langle \sigma_{Class}, \sigma_{Att}, \sigma_{Assoc}, \sigma_{Play} \rangle.$$

For each class,

$$\sigma_{Class}(c) \subseteq I_{Class}(c), \qquad |\sigma_{Class}(c)| < \infty.$$

For every inherited or direct attribute $a : t_c \to t$,

$$\sigma_{Att}(a) : \sigma_{Class}(c) \to I(t).$$

For every association,

$$\sigma_{Assoc}(as) \subseteq I_{Assoc}(as), \qquad |\sigma_{Assoc}(as)| < \infty,$$

subject to its declared multiplicities.

For each direct Role generalization $r \prec_d r'$,

$$\sigma_{Play}(r, r') \subseteq \sigma_{Class}(r) \times \sigma_{Class}(r'),$$

with exactly one parent-role instance for each specialized-role instance:

$$\forall x \in \sigma_{Class}(r) : \big|\{y \in \sigma_{Class}(r') \mid (x, y) \in \sigma_{Play}(r, r')\}\big| = 1.$$

This is the formal counterpart of the paper's requirement that enacting a specialized role entails the corresponding parent role.

6 OCL Invariants and Admissible States

Let $Inv(\mathcal{A})$ denote the conjunction of structural well-formedness constraints and user-defined OCL invariants. The admissible ACL states are

$$\Sigma_A = \mathcal{A} = \{\sigma \mid \sigma \models Inv(\mathcal{A})\}.$$

This notation is used uniformly by the iStar and BPMN semantics.

7 State Expressions and Post-State Expressions

Let

$$BoolExpr(\mathcal{A})$$

be OCL Boolean expressions evaluated in one ACL state, and

$$BoolPostExpr(\mathcal{A})$$

be OCL postcondition expressions evaluated over a pair (σ, σ'), where post-state references (e.g. OCL `@pre` conventions) are interpreted accordingly. We write

$$\sigma \models \varphi \quad \text{and} \quad (\sigma, \sigma') \models \psi.$$

Every ACL state participating in a system execution must remain admissible:

$$\sigma_i \in \Sigma_A \quad \text{for all execution steps } i.$$